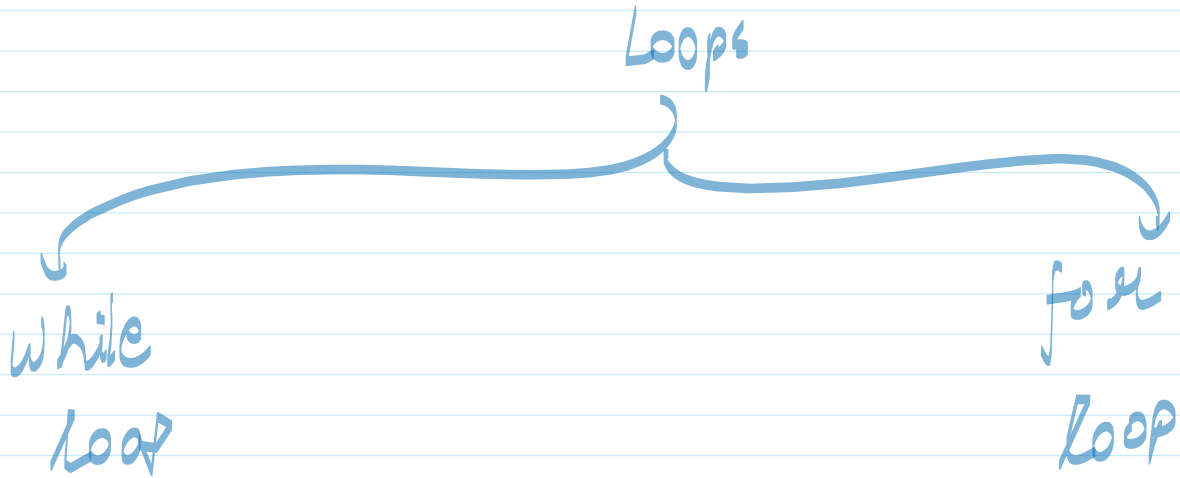


Loops :-

→ Wake up
→ Brush
→ Eat Breakfast

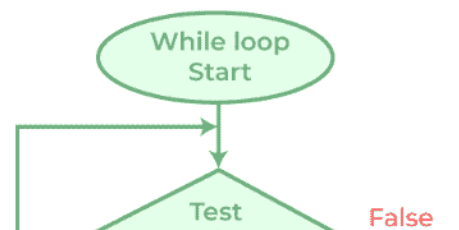
Repeats Everyday

loop



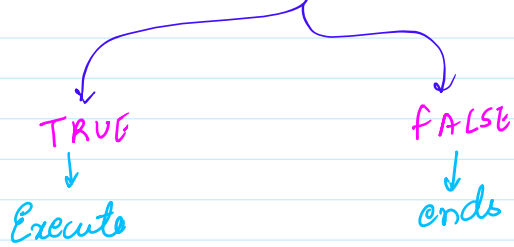
While loop :-

- 1) Loop start (Initialization)
- 2) Condition checks



1) Loop starts (Initialization)

2) Condition Checks



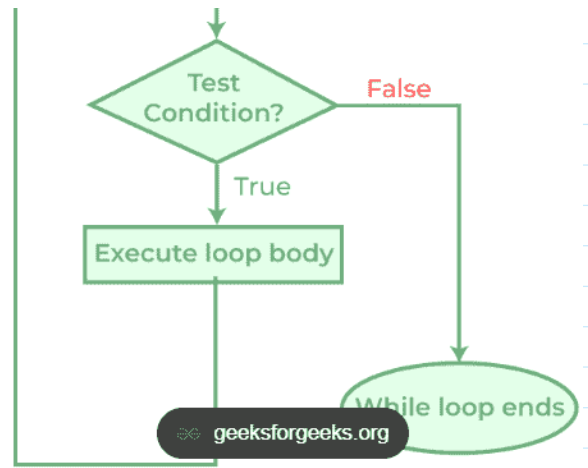
```
i=1 # Loop Initialization  
while i<=1000: i<=1000 → Condition  
    print("Sophie")  
    i+=1
```

loop body

$i = 1 \leq 1000$

$2 \leq 1000 \rightarrow \text{True}$

$1001 \leq 1000 \rightarrow \text{false}$



1. Sophie

2. Sophie

3. Sophie

Iteration:-

➤ One cycle of successful loop is called iteration.

While loop with Else:-

```
condition=False  
while condition==True:  
    print("Sophie")  
else:  
    print("Azaan")
```

Range function :-

range(start : stop : step)

range(0, 5): 0 1 2 3 4

range(1, 11): 1 2 3 4 5 6 7 8 9 10

Observation: Last no. does not count

For loop :-

```
for i in range(1, 1001):  
    print("Sophie")
```

i=1 # Loop Initialization

while i <= 1000:

print("Sophie")

i += 1

automatic

```
for i in range(1, 11, 2):  
    print("Sophie")
```

0

✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
1 2 3 4 5 6 7 8 9 10
i ↗ ↘ ↗ ↘ ↗ ↘ ↗ ↘ ↗ ↘

Sum of N natural number :

Ex- $1 + 2 + 3 + 4 + 5 + \dots + N = \frac{N(N+1)}{2}$

$N = 5$

$Sum = 0$ ← initially -

while